Internet Appendix
Keeping Options Open: What Motivates Entrepreneurs?

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A Model

A.1 Numerical solution

This section explains how the state and control variables of the problem are discretized on a grid and how the Bellman equation of the problem is solved numerically.

Discretization – The 5 state space and 2 control variables of the problem are discretized as follows. Financial wealth (W) is normalized by the average national wage and takes values between 0.01 and 500. The grid is uniformly distributed between the logs of these two bounds with a mesh size of $\Delta(w) = 0.1$. The agent's belief regarding the log of his permanent productivity (μ) takes 51 uniformly distributed values between -2 and 5. Simulations show that the agent never chooses to become/remain an entrepreneur if $\mu < -2$. Uncertainty regarding permanent productivity is a deterministic function of firm age, which is discretized using 41 points between 0 and 40 year old. I add three negative points to that grid: -3 denotes a former entrepreneur, while -2 and -1 respectively represent paid employees without and with an entrepreneurial idea, respectively. The persistent component of wages is discretized with 21 points uniformly distributed between -3 and 3 standard deviations using Tauchen's method. The transitory component is similarly discretized with 11 points.

Consumption is determined by the choice of the subsequent value of wealth. Capital is discretized with 41 points as a fraction of the maximum investment the agent can make: $k_{\max}$. This maximum takes the minimum of two values. The first one is the optimal capital for a risk-neutral and unconstrained agent. The second one is the largest investment that the true agent can make without taking the risk of being unable to consume at the end of the period, which would result in a utility of $-\infty$. Hence, the agent must have enough wealth to meet his liabilities if $Z = 0$. Therefore, the value of $k_{\max}$ is:

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$$K_{\max}(W,\mu,\sigma_\mu) = \min\left\{\left(\frac{\alpha\mathbb{E}[Z]}{\delta+r_D}\right)^{\frac{1}{1-\alpha}}, \frac{1+(1-\tau)r_D}{(r_D+\delta)(1-\tau)}W\right\} \tag{A.1}$$

where $\mathbb{E}[Z]$ is the expected value of Z from the agent's point of view, that is $\mathbb{E}[Z] = e^{\mu+\frac{\sigma_\mu^2+\sigma_{z,1}^2+\sigma_{z,2}^2}{2}}$.

Resolution of the Bellman equation – The model is solved backward by dynamic programming. Each year is split into two sub-periods.

a. In the second sub-period, the agent chooses her consumption and then receives labor income shocks and entrepreneurial ideas (on the first day of the following year). Using an apostrophe sign (′) to denote evolving state variables, the Bellman equation for this subperiod is:

$$\begin{cases} V_{t+\frac{1}{2}}(W,\mu,a,l_1,\mathbf{1}_{K>0}) & = \max\limits_{C}\left\{\frac{(C+\mathbf{1}_{K>0}B)^{1-\gamma}}{1-\gamma} + \beta\mathbb{E}V_{t+1}(W',a',\sigma_\mu',l_1',l_2')\right\} \\ \text{with} & W' = W - C \\ & l_1' \sim \mathcal{N}\left(\rho l_1, \sigma_{l,1}^2\right) \\ & l_2' \sim \mathcal{N}\left(0, \sigma_{l,2}^2\right) \\ \text{and if } a=-2 \text{ and with probability } p\text{:} & \\ & a' = a+1 \\ & \mu' \sim \mathcal{N}\left(\bar{z}, \sigma_{\nu,1}^2\right) \\ \text{else:} & \mu' = \mu \\ & a' = a \end{cases} \tag{A.2}$$

where a if firm age, C is chosen so that W' is on the grid and $a=-2$ indicates that the agent has not yet had an entrepreneurial idea.

b. In the first sub-period, the agent chooses his occupation and how much to invest in his firm if he chooses to be an entrepreneur. The Bellman equation for this subperiod is:

$$\begin{cases} V_t(W,\mu,a,l_1,l_2) & = \max\limits_{K}\left\{\mathbb{E}V_{t+\frac{1}{2}}(W',\mu',a',l_1)\right\} \\ \text{with, if } K>0\text{:} & W' = W + (1-\tau)\left(e^z K^\alpha - \delta K\right) + r(W-K) \\ & \mu' = \mu + \frac{\sigma_\mu^2+\sigma_{z,1}^2}{\sigma_\mu^2+\sigma_{z,1}^2+\sigma_{z,2}^2}\left(z-\mu\right) \\ & a' = a+1 \\ & z \sim \mathcal{N}\left(\mu, \sigma_\mu^2+\sigma_{z,1}^2+\sigma_{z,2}^2\right) \\ \text{and, if } K=0\text{:} & W' = (1+r)W + e^{f(t)+l_1+l_2} \\ & \text{if } a\geq 0,\ a'=-3 \text{ else } a'=a \\ & \mu' = \mu \\ \text{s.t.} & K=0 \text{ if } a-3 \text{ or } a=-2 \end{cases} \tag{A.3}$$

where σ_μ is a deterministic function of firm age a (see Equation (7)). I discretize z using 41 points on the interval $\mu \pm 3 \times \sqrt{\sigma_\mu^2 + \sigma_{z,1}^2 + \sigma_{z,2}^2}$. As there is no reason to expect W' and μ' to be on the grid, I estimate the continuation value $V_{t+\frac{1}{2}}(W', \mu', a', l_1)$ by interpolation. I interpolate the value function as follows. First, I multiply V_t by $1-\gamma$ if $\gamma > 0$ and take the log. I interpolate the resulting function with respect to log wealth and μ'. Finally, I take the exponential of the result and divide it by $1-\gamma$ if $\gamma > 1$. I find the optimal K in two steps. First, I assume that the agent chooses to become entrepreneur (if he can) and solves for the optimal $K > 0$. Under this assumption, l_2 is no longer a state variable, which reduces the computational cost of solving for K^*. I compute the optimal continuation value for $K^* > 0$ and compare it to the continuation value for $K = 0$. The agent chooses entrepreneurship if the former is above the latter.

Simulation – I save the optimal policies from 25 to 65 years old and then simulate the model. I initialize wealth at $W = 0.1$ and generate a random distribution for the persistent component of labor income using its stationary distribution, that is $\mathcal{N}\left(0, \frac{\sigma_{l,1}^2}{1-\rho^2}\right)$. The simulation follows the timeline of Figure 2. Stochastic processes are not discretized and optimal policies therefore need to be interpolated. This is relatively straightforward for investment and consumption but not for the occupation choice, which is a binary decision. While solving the Bellman equation, I save the ratio of the continuation values of entrepreneurship and paid employment. If $\gamma > 1$ ($\gamma \leq 1$), the agent is better off as an entrepreneur if this ratio is below (above) one. I use linear interpolation to estimate this ratio for the current coordinates of the agent.

A.2 SMM procedure

Global optimization routine – The SMM solves Equation (13) with respect to seven parameters: B, $\sigma_{\nu,1}$, $\sigma_{\nu,2}$, $\sigma_{z,1}$, $\sigma_{z,2}$, p, $\bar{z}$. This is done using the following procedure, inspired by the TikTak algorithm (Arnoud et al., 2019).

1. I start by delimiting the parameter space by assuming that $B \in [0; 2]$, $\sigma_{\nu,1} \in [0; 1]$, $\sigma_{\nu,2} \in [0; 1]$, $\sigma_{z,1} \in [0; 0.3]$, $\sigma_{z,2} \in [0; 0.4]$, $p \in [0; 0.1]$ and $\bar{z} \in [-0.5; 0.5]$.
2. I draw a low-discrepancy (Halton) sequence of N points θ_i within that space. Then, from $i = 1$ to $i = N$, I run the following procedure:
   (a) If $i > 1$, then $\tilde{\theta}_i = \theta_i x^i + (1 - x^i)\theta^*$, otherwise $\tilde{\theta}_i = \theta_i$.
   (b) Using $\tilde{\theta}_i$ as a starting point, run a local optimization to minimize the SMM objective function. I use MATLAB's implementation of the Nelder-Mead simplex algorithm.
   (c) If the result of the optimization is below θ^*, then replace θ^* with the new best.
   (d) If $i < N$, go back to (a). Otherwise, θ^* is the solution and the algorithm stops.

The idea of the algorithm is to run local optimizations using a large number of starting points. As the algorithm progresses, new starting points get closer and closer to the best solution found so far. The speed of convergence depends on x, which I set to 0.95. I allow 250 evaluations of the objective function during local estimation steps and set $N = 100$. This means that the model is solved approximately 25,000 times by SMM. Solving the model takes approximately 2 minutes on a high-end graphic card (Nvidia Tesla K80). In practice, I find that $N = 10$ is sufficient to get estimates very close to the final solution. Almost all local minimization steps converge to the same region and exceptions return much higher error functions. This makes me confident that the procedure converges to the global minimum.

Standard errors – Standard errors are estimated as the square roots of the diagonal elements of matrix Q, which is defined as:

$$Q = [J^T \boldsymbol{\Omega}^{-1} J]^{-1},$$

where $\boldsymbol{\Omega}$ is the adjusted empirical variance-covariance matrix of targeted moments and J is the Jacobian matrix of the simulated moments with respect to estimated parameters. Because this SMM estimation uses plug-in parameters $(\rho, \sigma_{l,1}^2, \sigma_{l,2}^2)$ estimated directly from income data moments (v_1, v_3, v_5), the matrix $\boldsymbol{\Omega}$ contains an adjustment for the errors in the first stage of the estimation. Specifically,

$$\boldsymbol{\Omega} = \boldsymbol{\Omega}^{\text{no adj}} + \left[\frac{\partial \hat{\boldsymbol{m}}}{\partial \boldsymbol{v}}\right] \boldsymbol{\Xi} \left[\frac{\partial \hat{\boldsymbol{m}}}{\partial \boldsymbol{v}}\right]^T,$$

where $\boldsymbol{\Omega}^{\text{no adj}}$ is the variance-covariance matrix of targeted moments in the second stage (i.e., ignoring the first-stage errors), $[\partial \hat{\boldsymbol{m}}/\partial \boldsymbol{v}]$ is the Jacobian matrix of moments with respect to the variances $\boldsymbol{v} = [v_1, v_3, v_5]^T$, and $\boldsymbol{\Xi}$ is the empirical variance-covariance matrix of $\boldsymbol{v}$.[1] The Jacobian matrices J and $[\partial \hat{\boldsymbol{m}}/\partial \boldsymbol{v}]$ are approximated around the SMM estimate using forward finite differences with a spacing of 0.01.

[1]This expression is technically an approximation of the fully adjusted matrix, which adds negligible cross-product terms.

B Additional tables

Table B.1: **Matching Quality**

This table reports the balance between the sample of entrepreneurs and the benchmark group. Each new entrepreneur is matched with an individual having the same previous occupation, gender and labor force participation status over the previous three years. The final match minimizes the Mahalanobis distance in terms of age and earnings for each of the three years preceding the entrepreneurial spell.

	Mean		Covariance matrices							
	Entrepreneurs	Benchmark	Entrepreneurs				Benchmark			
Age$_{t_0}$	40.49	40.35	78.5				72.6			
Income$_{t_0-1}$	1.439	1.447	3.48	2.13			3.61	1.99		
Income$_{t_0-2}$	1.588	1.598	4.02	1.74	2.01		4.03	1.71	1.90	
Income$_{t_0-3}$	1.610	1.626	4.46	1.61	1.78	2.05	4.69	1.58	1.72	1.87

Table B.2: **Earnings and Exit rate Differences between Samples**

This table reports earnings and exit rates for firms that hired at least one employee in the first year. In Panel A, the sample is restricted to firms who paid their owner a wage in the first two years whereas Panel B takes into account the full sample.

	Age of firm									
	1	2	3	4	5	6	7	8	9	10
Panel A: At least one employee at creation and early wage to entrepreneur										
Net Income + Owner's Wage (k€)	12.9	16.6	19.5	23.3	25.9	28.3	27.9	26.6	30.4	32.6
	(1.0)	(1.0)	(1.1)	(1.2)	(1.2)	(1.3)	(1.4)	(1.4)	(1.5)	(1.6)
Exit rate (%)	5.8	8.1	8.6	7.9	7.3	6.6	6.1	6.1	6.2	5.9
	(0.1)	(0.1)	(0.1)	(0.1)	(0.1)	(0.1)	(0.1)	(0.1)	(0.1)	(0.1)
Panel B: At least one employee at creation										
Net Income + Owner's Wage (k€)	10.1	13.3	16.8	20.1	23.5	26.0	26.2	28.2	30.4	32.5
	(0.4)	(0.4)	(0.4)	(0.4)	(0.5)	(0.5)	(0.5)	(0.6)	(0.6)	(0.7)
Exit rate (%)	7.3	9.7	9.3	8.7	8.2	7.4	6.9	6.8	6.9	6.5
	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)

Table B.3: **Sensitivity of Parameter Estimates to Risk Aversion**

This table reports the sensitivity of my baseline parameter estimates to the coefficient of relative risk aversion.

		(1)	(2)	(3)	(4)
		$\gamma = 0.5$	$\gamma = 1$	$\gamma = 1.5$	$\gamma = 2$
B	Unobserved benefit	.330	.386	.395	.402
		(.003)	(.003)	(.002)	(.003)
$\sigma_{\nu,1}$	Dispersion of priors	.811	.696	.712	.630
		(.001)	(.002)	(.002)	(.002)
$\sigma_{\nu,2}$	Initial uncertainty	.294	.485	.477	.512
		(.002)	(.002)	(.002)	(.001)
$\sigma_{z,1}$	SD of permanent shocks	.180	.183	.189	.182
		(.000)	(.000)	(.000)	(.000)
$\sigma_{z,2}$	SD of transitory shocks	.276	.283	.279	.280
		(.000)	(.000)	(.000)	(.000)
p	Arrival rate of ideas	.043	.036	.039	.043
		(.000)	(.000)	(.000)	(.000)
$\bar{z}$	Mean log productivity	-.252	-.217	-.246	-.191
		(.002)	(.002)	(.002)	(.003)
α	Returns to scale	.260	.260	.260	.260
		(.001)	(.001)	(.001)	(.002)

Table B.4: **Counterfactual Experiments – Robustness Checks**

This table reports the same counterfactual experiments as Table 4. In Panels I to III, I assume different model specifications, for which estimated parameters are reported in Table 3.

		Removed Element				
	(1)	(2)	(3)	(4)	(5)	(6)
		TFP Shocks	Uncertainty	Benefits	Option	Combined
Panel I: Labor Market Stigma						
I.A: Changes in Entrepreneurial Sector						
Firm creations		-4.2%	-12.6%	-17.2%	-19.7%	-39.4%
Firms		-8.0%	-0.4%	-25.2%	+0.4%	-35.2%
Mean Value Added		-36.2%	+2.1%	+22.7%	-11.4%	+12.3%
Total Value Added		-41.3%	+1.6%	-8.2%	-11.0%	-27.2%
I.B: Entrepreneurial Premium over first 10 years						
All entrepreneurs	38.2%	31.3%	56.0%	54.7%	64.6%	105.5%
Active entrepreneurs	57.4%	47.0%	65.8%	93.8%	64.6%	105.5%
Panel II: Liquidation Costs						
II.A: Changes in Entrepreneurial Sector						
Firm creations		-3.7%	-9.1%	-13.9%	-20.9%	-39.6%
Firms		-8.5%	-0.2%	-24.0%	+3.3%	-33.2%
Mean Value Added		-41.2%	+1.7%	+21.2%	-12.7%	+9.7%
Total Value Added		-46.2%	+1.4%	-7.9%	-9.9%	-26.7%
II.B: Entrepreneurial Premium over first 10 years						
All entrepreneurs	36.9%	28.2%	47.7%	49.2%	61.0%	100.5%
Active entrepreneurs	51.9%	39.1%	57.8%	84.3%	61.0%	100.5%
Panel III: Progressive Income Tax						
III.A: Changes in Entrepreneurial Sector						
Firm creations		-1.6%	-6.8%	-11.3%	-17.0%	-40.4%
Firms		-5.3%	-0.0%	-28.4%	+8.1%	-38.6%
Mean Value Added		-40.1%	+1.1%	+25.9%	-15.0%	+13.5%
Total Value Added		-43.3%	+1.1%	-9.8%	-8.1%	-30.3%
III.B: Entrepreneurial Premium over first 10 years						
All entrepreneurs	36.7%	26.1%	45.0%	48.7%	51.7%	103.7%
Active entrepreneurs	49.9%	34.9%	54.0%	87.2%	51.7%	103.7%

Table B.4 *continued*

In Panels IV to VI, I assume different levels of relative risk aversion, for which estimated parameters are reported in Table B.3.

		Removed Element				
	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	Permanent TFP Shocks	Initial Uncertainty	Unobserved Benefits	Exit Option	(4) & (5) Combined
Panel IV: Collateral Constraint						
IV.A: Changes in Entrepreneurial Sector						
Firm creations		-1.3%	-6.0%	-12.9%	-17.1%	-41.3%
Firms		-5.5%	-0.0%	-30.2%	+5.3%	-40.3%
Mean Value Added		-32.4%	+0.7%	+27.8%	-13.0%	+18.1%
Total Value Added		-36.1%	+0.6%	-10.8%	-8.3%	-29.5%
IV.B: Entrepreneurial Premium over first 10 years						
All entrepreneurs	35.7%	28.0%	43.5%	47.3%	49.5%	101.3%
Active entrepreneurs	48.1%	37.3%	52.3%	86.3%	49.5%	101.3%
Panel V: Relative Risk Aversion of $\gamma = 0.5$						
IV.A: Changes in Entrepreneurial Sector						
Firm creations		-2.1%	-2.6%	-16.8%	-20.3%	-40.8%
Firms		-9.7%	+0.1%	-23.0%	+8.9%	-21.1%
Mean Value Added		-38.0%	+0.2%	+20.9%	-10.6%	+10.0%
Total Value Added		-44.1%	-0.3%	-6.9%	-2.5%	-13.3%
IV.B: Entrepreneurial Premium over first 10 years						
All entrepreneurs	37.4%	29.2%	40.0%	53.8%	61.4%	101.3%
Active entrepreneurs	50.2%	39.1%	52.2%	83.1%	61.4%	101.3%
Panel VI: Relative Risk Aversion of $\gamma = 2$						
VI.A: Changes in Entrepreneurial Sector						
Firm creations		-1.7%	-7.8%	-7.2%	-23.1%	-41.7%
Firms		-3.0%	+0.1%	-26.9%	-5.7%	-56.1%
Mean Value Added		-40.7%	+2.4%	+22.0%	-12.4%	+14.3%
Total Value Added		-42.5%	+2.5%	-10.8%	-17.4%	-49.9%
VI.B: Entrepreneurial Premium over first 10 years						
All entrepreneurs	37.3%	25.8%	48.0%	45.8%	69.7%	117.5%
Active entrepreneurs	51.4%	35.1%	58.6%	83.6%	69.7%	117.5%

Table B.4 *continued*

	(1) Baseline	(2) Permanent Shocks	(3) Initial Uncertainty	(4) Unobserved Benefits	(5) Exit Option	(6) (4) & (5) Combined
		Removed Element				
Panel VII: Heterogenous preferences						
Panel A: Changes in Entrepreneurial Sector						
Firm creations		-2.7%	-4.9%	-14.8%	-19.3%	-42.3%
Firms		-8.6%	-0.1%	-25.3%	+5.8%	-33.2%
Mean Value Added		-41.9%	+0.4%	+21.5%	-12.6%	+12.6%
Total Value Added		-46.9%	+0.3%	-9.2%	-7.6%	-24.8%
Panel B: Premium over first 10 years						
All entrepreneurs	36.7%	25.4%	44.9%	50.8%	57.4%	103.4%
Active entrepreneurs	50.2%	32.5%	55.8%	84.7%	57.4%	103.4%

C Additional figures

Figure C.1: **SMM Fitness without key parameters**

This figure reports all empirical moments targeted in my structural estimation as well as their simulated counterparts in the model either without unobserved benefits ($B = 0$), or without permanent productivity shocks ($\sigma_{z,1} = 0$) or without initial uncertainty ($\sigma_{\nu,2} = 0$). In each case, I reestimate the model by SMM using the same set of empirical moments.

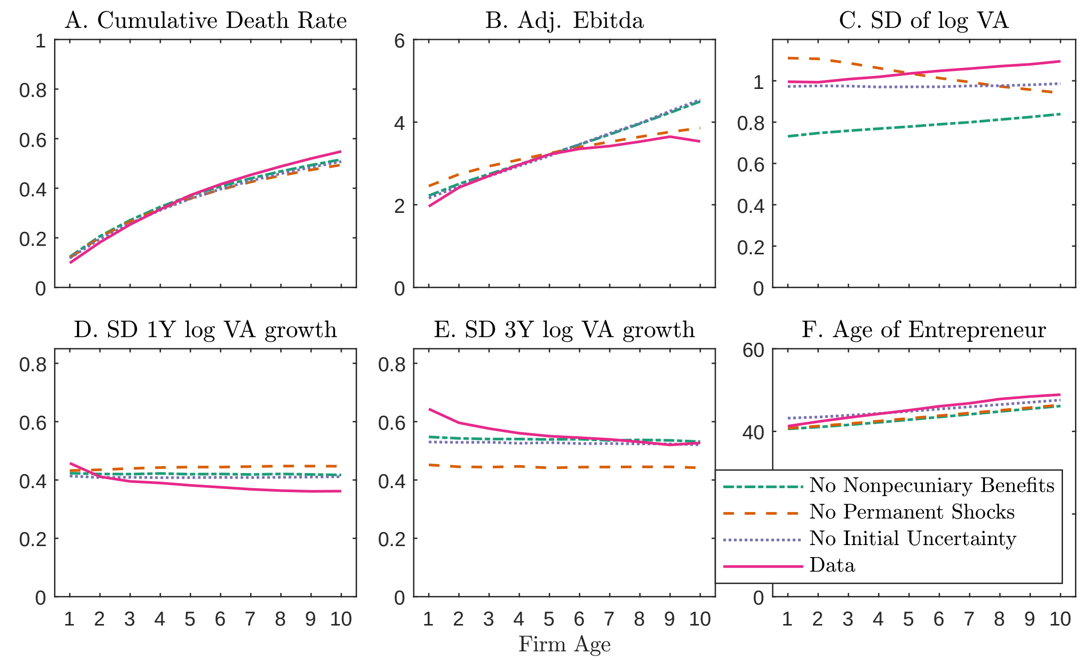


Figure C.2: **Bias of Matching Method in Simulated Data**

This graph reports the mean income of individuals becoming entrepreneurs, that of a benchmark group constructed by matching, and the true wage that entrepreneurs would have received as paid employees. The model is simulated using estimated parameters reported in Column (1) of Table 3. The simulated matching procedure is based on the last three years of wages.

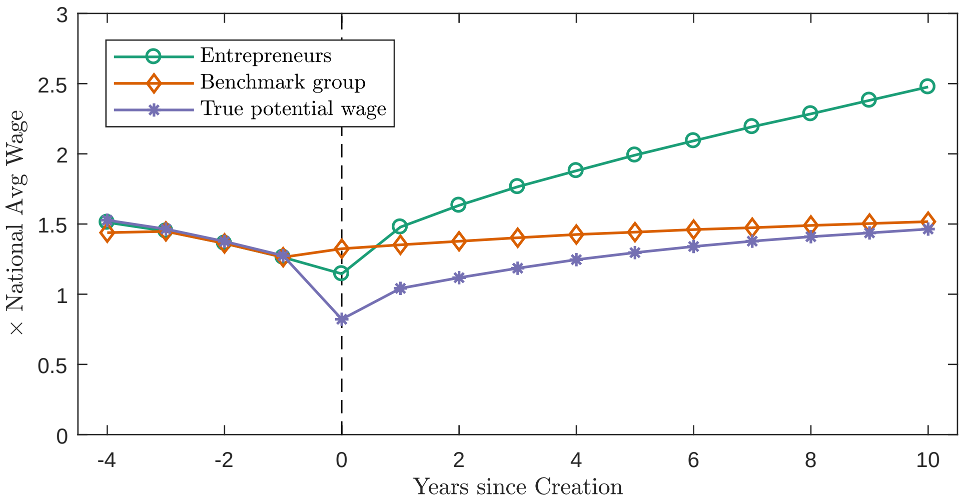


Figure C.3: **Income Trajectories for Short Entrepreneurial Spells (<3 years)**

This graph reports the income trajectories of entrepreneurs closing their firms in the first three years and that of a benchmark group built by matching. The mean income of entrepreneurs is defined as the sum of wages and net income. The matching algorithm takes into account gender, pre-entrepreneurial occupation and wages received over the last three years.

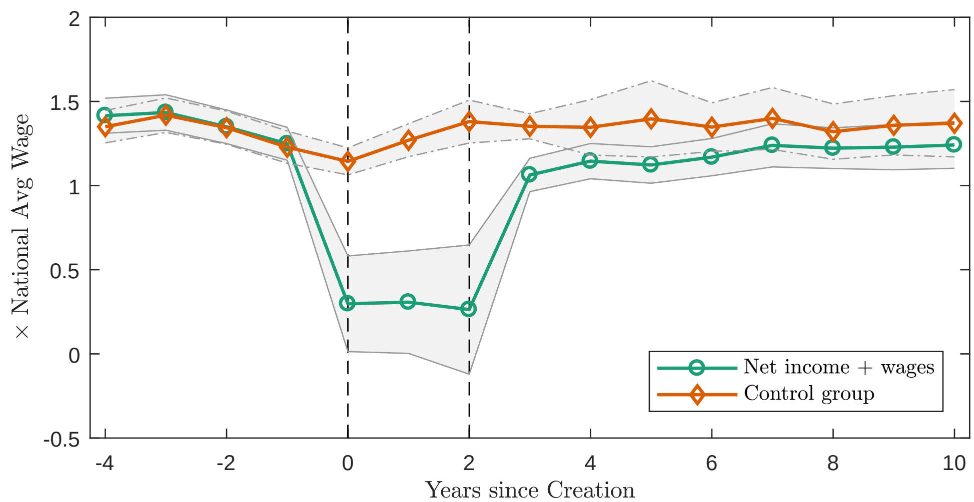


Figure C.4: **Income Trajectories for Short Entrepreneurial Spells (<3 years) relative to all Entrepreneurs**

This graph reports the income trajectories of entrepreneurs closing their firms in the first three years and that of all entrepreneurs. The mean income of entrepreneurs is defined as the sum of wages and net income.

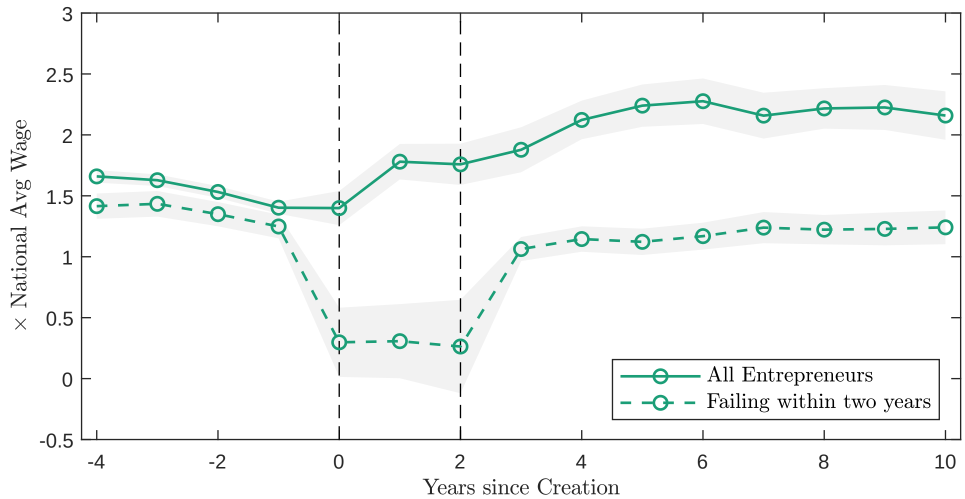
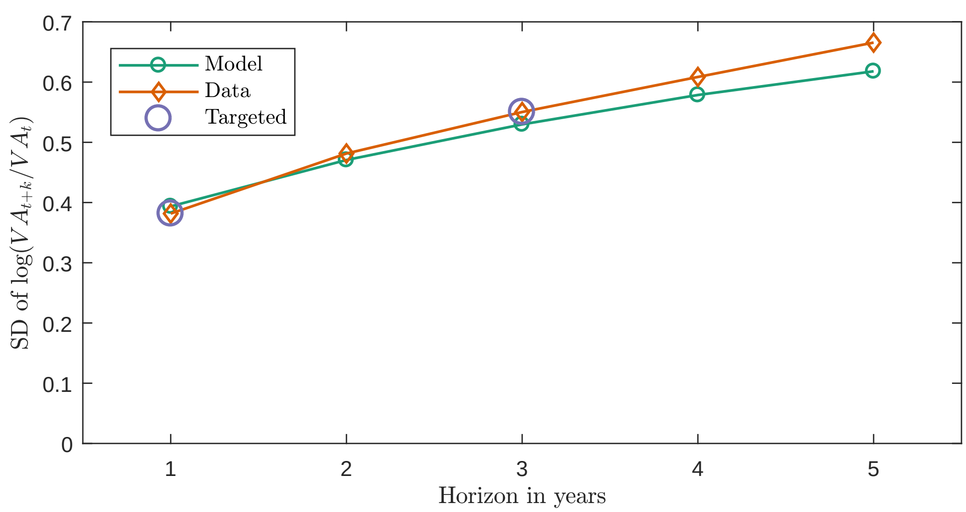


Figure C.5: **Validity of Ramdom Walk Assumption**

This figure reports the standard deviation of log growth rates at different time horizons for $t = 5$ year old firms. Growth rates are defined as $\log \mathrm{VA}_{it+k} - \log \mathrm{VA}_{it}$, where k is the time horizon in years. Simulated data are generated using my baseline specification and parameters reported in Column (1) of Table 3.



References

Arnoud, Antoine, Fatih Guvenen, and Tatjana Kleineberg, "Benchmarking Global Optimizers," *Unpublished working paper*, October 2019, (26340).